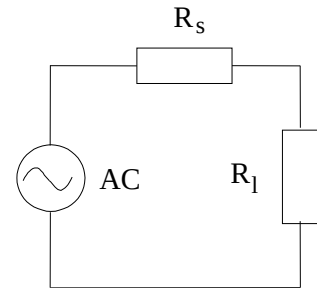


Problem Sheet 2

1.

(a) For a real (resistive) source impedance R_s and load R_L show that maximum power is transferred from source to load when $R_s = R_L$.

(b) Show that the efficiency η is given by $\eta = \frac{R_L}{R_L + R_s}$.



2.

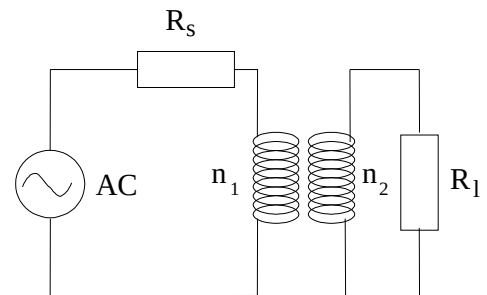
(a) An AC voltage source has an output resistance R_s and produces a signal of amplitude V_0 (root mean square, or RMS value). The source is used to drive a load with resistance R_L . For $V_0 = 28 \text{ V}_{\text{RMS}}$, $R_s = 500\Omega$, $R_L = 8\Omega$, how much power is dissipated in R_s and R_L , and what is the efficiency?

(b) Show that to optimise power transfer to the load we can add a signal transformer (as in the figure) as long as:

$$\frac{n_1}{n_2} = \sqrt{\frac{R_s}{R_L}},$$

where n_1 and n_2 are the numbers of turns in the primary and secondary sides of the transformer.

(c) For this 'matched' case what is the power now dissipated in R_s and R_L ? What is the efficiency?



3.

(a) In the case of an electrical signal travelling from a medium with characteristic impedance Z_1 into a medium with characteristic impedance Z_2 , state the boundary conditions for voltage and current. What is the physical reasoning which leads us to these boundary conditions?

(b) Since we can relate the voltages and currents by $V = IZ$, find the amplitude reflection coefficient k_r , which is defined as the ratio of the reflected signal to the incident signal amplitude.

(c) Calculate the amplitude transmission coefficient k_t , the ratio of the transmitted to incident amplitudes.

4. SEMINAR QUESTION (PART I)

(a) A 200 m long coaxial cable is fabricated with a capacitance per unit length of 100 pF/m and an inductance per unit length of $5 \times 10^{-7} \text{ H/m}$. A single 10 ns FWHM Gaussian voltage pulse with amplitude of 10 volts is introduced into one end of the cable. The input to the cable is terminated with a 10Ω resistor, the output with a 100Ω resistor. What is the characteristic impedance of 10, 20 and 30 m lengths of this cable, and at what speed do electrical signals propagate in the cable?

(b) If it were possible to fabricate a cable with the same inductance per metre, but ten times less capacitance per metre, what would the signal velocity now be? Take care over whether your answer is physically achievable or not.

Instrumentation

(c) How much signal is reflected from the output of the cable on the first transit of the pulse through this transmission line? Is the reflected signal inverted or un-inverted?

(d) How long does it take for the first and second *reflections* of the voltage pulse to reach the output end of the cable? What are the amplitudes and signs of these pulses?

5.

(a) An antenna with an impedance of 100Ω needs to be connected via a 50Ω cable into a receiver (also 50Ω). To transform the impedance, a short intermediate-section of cable is used. What should be its impedance, and where should it be placed?

(b) To ideally match a signal received at 141 MHz, how long should the section be?

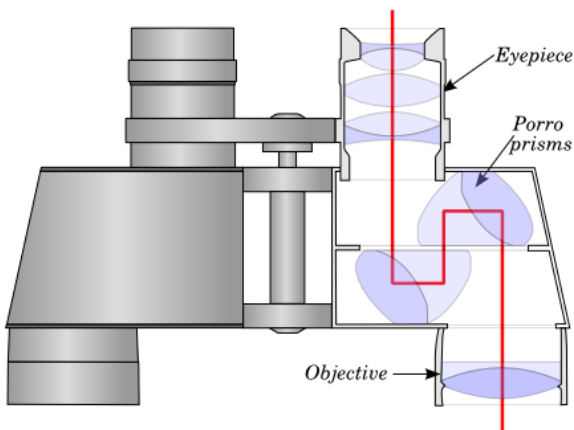
6. SEMINAR QUESTION (PART II)

[Note: this specific optics problem would previously have been covered in lectures – but you should still be able to solve it essentially from your understanding of transmission lines; as Feynman was fond of saying, “the same equations have the same solutions”. You may want to recall that the characteristic impedance of an optical medium is $Z_{em} = \sqrt{\mu\mu_0/(\epsilon\epsilon_0)}$, the speed of light in the medium is $c = 1/\sqrt{\mu\mu_0 \epsilon\epsilon_0}$, and the refractive index is $n = \sqrt{\epsilon\mu} \cong \sqrt{\epsilon}$ (since $\mu \approx 1$ for non-magnetic materials).]

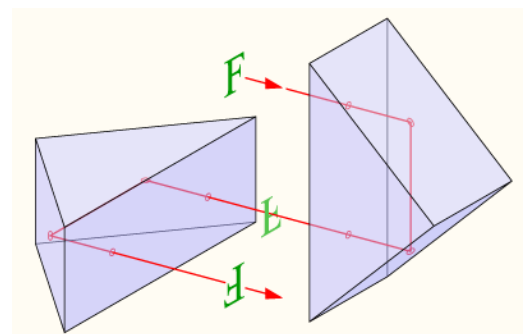
The arrangement of optical elements in a pair of high-quality binoculars is shown in the figure below. The objective lens collects light and brings it to focus creating a real image at a distance equal to the focal length behind the lens. The eyepiece is used to magnify this real image which is then seen as a virtual image by the eye.

Between objective and eyepiece is a pair of Porro-prisms (below) which serve 3 functions:

1. They invert the image, which would otherwise be seen upside down by the eye;
2. They increase the optical path length of the binoculars;
3. They widen the separation of the objective lenses, thus enhancing the stereo-image.



The 45° incidence angle of the light means that total internal reflection occurs at each reflection, with negligible loss of signal. High-quality optics usually uses multiple lenses at the eyepiece to correct for the physical effects of spherical and chromatic aberration inherent in even the best quality lenses. Three of the lenses are actually “doublets” composed of two glasses of different refractive indices bonded together to further correct for these aberrations. In these binoculars, the concave lenses and Porro-prisms are made of glass with $n_1 = 1.52$ and the convex lenses have $n_2 = 1.75$.



(a) Assuming uncoated optics, and given that the refractive index of air is 1, calculate the reduction in light intensity through the binoculars.

(b) To combat this problem, coated optics are often used. These employ a layer of material with an intermediate refractive index, and it can be shown that the index for minimum reflection is given by:

Instrumentation

$$n_{\text{coating}} = \sqrt{n_{\text{air}} \times n_{\text{glass}}}$$

It is not usually possible to exactly match this precisely; however, the material magnesium fluoride (MgF_2) is often used since it has a refractive index of 1.38. For a single convex lens as used above, how much is the reflected intensity reduced?

(c) What thickness of coating would be required for perfect interference cancellation of green light roughly in the centre of the visible spectrum at $\nu = 560 \times 10^{12}$ Hz.

7.

(a) A length of co-axial cable with characteristic impedance 50Ω is cut in two and the halves are connected together in parallel. What is the characteristic impedance of the parallel combination, and why?

(b) An electrical pulse travelling down a transmission line is described as having a Gaussian shape:

$$V = Ae^{-\frac{x^2}{2\sigma^2}}$$

Show that the Full-Width Half Maximum (FWHM) of the pulse is roughly 2.35σ .

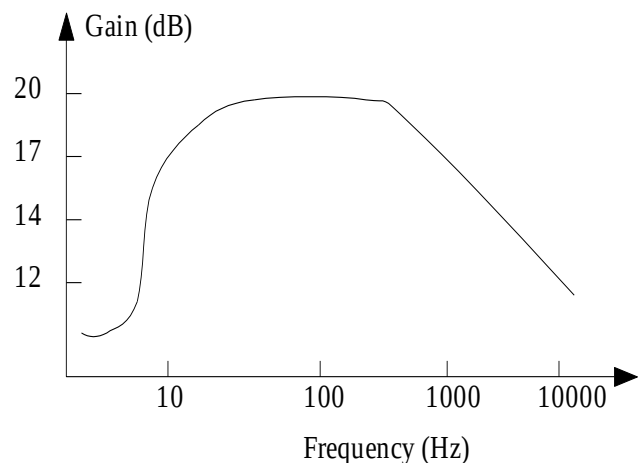
(c) Why is not physically realistic for the pulse to be *truly* Gaussian in shape?

8.

(a) An amplifier is set up as an AC-coupled voltage amplifier, which means that it has zero gain for a DC input. At high-frequencies, it exhibits a natural fall-off of the gain. Its performance in this configuration is described by the Bode plot shown (magnitude term only). What is the voltage gain of the system (in real numbers rather than dB) at 100 Hz?

(b) What is the bandwidth of the system?

(c) What is the high-frequency roll-off, expressed in dB per decade of frequency?



9.

(a) An analogue signal has a maximum amplitude of 1 V, a minimum amplitude of 0 V. The signal is digitized with an 8-bit converter. What voltage is represented by the digital numbers 10 and 255? What is the resolution R of the digitization?

(b) The signal has a random noise of 1mV, this being the standard deviation of the noise component of the signal. What is the noise in the digitized signal expressed in terms of R ?

(c) Since the quantisation noise is evenly distributed between $\pm R/2$, express the quantisation noise in terms of R and hence calculate the total noise in the digital signal. By what percentage does the quantisation process increase the noise compared to the original signal?

Instrumentation

Numerical Answers and Hints

2. (a) 1.5W, 24mW, 1.6% (c) 392mW, 50%.
4. (a) 70.7Ω , $0.47c_0$ (c) +1.72V (d) $4.26\mu s$ and -1.29V, $7.09\mu s$ and +0.17V.
5. (a) 70.7Ω (b) 25cm.
6. Assume that the relative permeability of glass is 1 (it is a non- magnetic material). Light intensity is proportional to amplitude squared.
(a) 54% (b) to 53% of the un-coated value (c) 97 nm.
7. (a) Consider the effect adding inductance and capacitance in parallel will have on C' and L' (c) Will there be zero volts anywhere on the line?
8. (a) 10 (b) 990 Hz (c) -5 dB/decade.
9. Assume that the converter rounds to the nearest digital value
(a) $10R \pm \frac{R}{2}$ V and $255R \pm \frac{R}{2}$ V, $R = \frac{1}{256}V$ (b) 0.26R (c) 50%.